FINAL PROJECT REPORT

# NYC Airbnb Market Analysis

Market overview, pricing patterns, availability, reviews, and neighborhood performance

**DATA ANALYSIS • PYTHON • POWER BI**

## Executive Summary

This project analyzes Airbnb listings in New York City using the 2019 listings dataset. The analysis explores listing distribution by borough and room type, price differences, availability, review activity, and neighborhood concentration. Python supports cleaning and exploratory analysis; Power BI presents the results in an interactive two-page dashboard.

| 48,895 | $152.72 | 112.78 days | 1,138,005 |
|---|---|---|---|
| Total listings | Average listed price | Average availability | Total reviews |

## Project Objectives

- Prepare a consistent dataset for analysis.
- Compare listing counts and average prices across boroughs.
- Explore room-type availability and price-tier distribution.
- Communicate actionable findings through an interactive dashboard.

## Deliverables

Cleaned CSV, Python analysis notebook, two-page Power BI dashboard, and this final report.

# 1. Dataset & Methodology

## Dataset overview

The original Airbnb NYC 2019 dataset contains 48,895 records and 16 source columns. The cleaned analysis table used in the project contains 48,895 rows and 19 columns, including additional flags for zero-price listings and IQR-based outliers in price and minimum nights.

## Data preparation

The workflow normalizes text fields, removes embedded line breaks and excess whitespace, standardizes missing listing/host labels where needed, parses last-review dates, and sets reviews-per-month to zero only for listings with no recorded reviews. Outlier flags are kept for investigation rather than assuming every extreme value is invalid.

## Analysis approach

Python is used for cleaning and exploratory analysis. Power BI visualizes KPIs, borough comparisons, room-type availability, price-tier mix, listing geography, neighborhood rankings, review-date trends, and host-level summaries.

## Metric definitions & caveats

| Metric | Definition / interpretation |
|---|---|
| Average Price | Mean of the listed price field; not a realized transaction price. |
| Average Availability | Average number of days available in the 365-day availability field. |
| Estimated Occupancy Rate | (365 – average availability) ÷ 365. This is only a proxy; unavailable days may be blocked rather than booked. |
| Active Listings | The dashboard measure counts listings with more than 50 reviews; this does not prove a listing is currently active. |
| Estimated Annual Revenue | A price × (365 – availability) proxy. It is not verified booking revenue; label it Estimated Revenue Proxy. |
| Price Tier | Budget (<100), Standard (100–300), and Luxury (>300) using the project's current rule. |

## Limitations

This is a 2019 snapshot, not a view of today's market. The source does not verify actual bookings, realized revenue, or whether a listing is currently bookable. The review-date line chart groups listings by their last review date; it is not a historical series of repeated price observations.

# 2. Key Findings

## Average listed price by borough

| Borough | Average price | Finding |
|---|---|---|
| Manhattan | $211.79 | Highest average listed price in the displayed comparison. |
| Brooklyn | $89.78 | Second-highest average listed price. |
| Queens | $70.13 | Lower average than Manhattan and Brooklyn. |
| Bronx | $58.57 | Lower average listed price than the first three boroughs. |
| Staten Island | $49.76 | Lowest average listed price in the displayed comparison. |

## Price-tier distribution

The dashboard shows approximately 24K Standard listings (48.39%), 22K Budget listings (44.74%), and 3K Luxury listings (6.87%). Standard and Budget together account for about 93.13% of listings under the project's thresholds.

## Availability by room type

Shared rooms show the highest average availability (about 160 days). Entire homes/apartments and private rooms are each around 110 days in the dashboard. Availability alone does not confirm occupancy or demand.

## Neighborhood concentration

The displayed Top 10 chart is led by Williamsburg (about 3.9K listings), Bedford-Stuyvesant (about 3.7K), and Harlem (about 2.7K). These high-supply neighborhoods are useful candidates for deeper comparison of prices, room types, and availability.

## Last-review-date trend

The line chart shows average price by year based on each listing's last-review date. It has a visible peak around 2013 and lower averages in later years. Interpret cautiously: this is not proof that Airbnb prices moved this way over time.

# 3. Business Insights & Recommendations

## Business Insights

**1. Manhattan commands a price premium.** Its average listed price ($211.79) is far higher than the other borough averages shown. Compare within borough by room type and neighborhood before making pricing decisions.

**2. Budget and Standard dominate the listing mix.** Together they represent about 93.13% of listings under the chosen thresholds, so competition and differentiation within these segments deserve attention.

**3. Supply is concentrated in a few neighborhoods.** Williamsburg, Bedford-Stuyvesant, and Harlem lead the displayed Top 10. High listing volume can indicate stronger competition; evaluate alongside price and availability.

**4. Availability differs by room type.** Shared rooms have higher average availability in this dataset. This may reflect different booking patterns, but it cannot independently prove lower demand.

**5. Reviews are context, not revenue.** Review counts are affected by listing age and guest review behavior. They should not be treated as direct booking counts or income.

**6. Occupancy and revenue values are proxies.** The 69.10% estimated occupancy and approximately 2bn revenue proxy are formula-based estimates, not verified transaction data.

## Recommended next steps

• Compare prices by borough, neighborhood, and room type together.
• Inspect price and minimum-night outliers before using them in decisions.
• Add dashboard notes that clearly identify occupancy and revenue as estimates.
• If booking-level data becomes available, replace proxy measures with booked nights and realized revenue.
• Label the review-date line chart carefully so users do not mistake it for historical price tracking.

# 4. Dashboard Summary

Page 1 — NYC Airbnb Market Overview: listing distribution by borough, geographic map, neighborhood ranking, and average price comparison.
Page 2 — NYC Airbnb Pricing & Performance: room-type availability, price-tier mix, average price by borough, last-review-date trend, and host summary table.

## Conclusion

The analysis shows clear geographic price differences, a listing mix dominated by Budget and Standard tiers, and supply concentration in several neighborhoods. These findings provide a useful basis for comparison, while occupancy, revenue, and historical price conclusions should remain cautious because the dataset does not include confirmed booking transactions or repeated price observations.

**Prepared for:** Voltix — Task 12
**Project:** Airbnb NYC Data Analysis
**Tools:** Python, pandas, Power BI